

José A. Benavides

Curriculum Vitae

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Education

- 2018 – 2023 **Ph.D. in Astronomy**, *National University of Córdoba, Argentina*
Thesis title: *Assembly of Galaxy Clusters and Formation of Ultra-Diffuse Galaxies*
- 2012 – 2015 **M.Sc. in Physics.**, *National University of Colombia, Colombia.*
Thesis title: *Numerical Simulations of the Infall of Dwarf Galaxies Associations Into the Milky Way Halo.*
- 2006 – 2012 **B.S. in Physics**, *National Pedagogical University, Bogotá - Colombia*

Research Experience

- 03/2026 – Present **Assistant Project Scientist**, *Physics & Astronomy Department, University of California, Riverside - United States*
- 03/2023 – 03/2026 **Postdoctoral Researcher**, *Physics & Astronomy Department, University of California, Riverside - United States*
- 05/2025 – 07/2025 **Visiting Researcher**, *Institute for Computational Cosmology (ICC) - Durham University, Durham - United Kingdom*
- 03/2022 – 06/2022 **Visiting Researcher - LACEGAL**, *Istituto Nazionale di Astrofisica (INAF) & Osservatorio Astronomico di Trieste, Italy*

Awards

- 04/2021 – 03/2023 **Doctoral Fellowship**, *Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET), Argentina*
- 04/2018 – 03/2021 **Doctoral Fellowship**, *Secretaría de Ciencia y Tecnología (SeCyT), UNC, Córdoba - Argentina*
- 04/2014 – 03/2015 **Young Resercher Fellowship**, *National University of Colombia, Faculty of Sciences, Bogotá - Colombia*
- 03/2013 – 03/2014 **Magister's Fellowship**, *National University of Colombia, Faculty of Sciences, Bogotá - Colombia*

Teaching Experience

- 02/2026 – Present **Physics Instructor (Part-Time)**, *Riverside City College, Riverside - California*
- 08/2020 – 03/2022 **Teaching Assistant**, *National University of Córdoba, FAMAf, Córdoba - Argentina*
- 08/2013 – 04/2018 **Assistant Professor (Full-time)**, *Antonio Nariño University, Faculty of Sciences, Bogotá - Colombia*
- 04/2013 – 12/2013 **Teaching Assistant**, *National University of Colombia, Faculty of Sciences, Bogotá - Colombia*
- 03/2012 – 03/2013 **Teacher, high school (Full-time)**, *Colegio del Bosque Bilingüe, UAN, Bogotá - Colombia*

Selected talks

- 06/2025 **Formation and Evolution of Ultra-Diffuse Galaxies in TNG50**, *Cosmology/Galaxy seminar, MPA, Munich*
- 05/2025 **Exploring some challenges in dwarf galaxies with cosmological simulations**, *TRACES meeting (invited talk), IAC, Tenerife*

- 05/2025 **Dwarf Galaxies and Open Challenges in LCDM: Morphology, UDGs, and Quenching in the Field**, *Durham Astronomy Colloquium*, ICC, Durham
- 11/2024 **Morphological transition in simulated low-mass galaxies in FIREbox**, *Galaxy group meeting/seminar*, CCA, Flatiron Institute, New York
- 04/2022 **Formation of Ultra-Diffuse galaxies in different environments**, *IFPU Special Seminar*, Trieste

Mentoring & Outreach Experience

- 2023–Present I am currently co-mentoring a PhD student at UCR whose first paper, on the *tidal track evolution of compact and ultra-faint galaxies*, is nearing submission
- 2022–2023 I advised a undergraduate student whose thesis on *Compact Groups of Galaxies* led to a publication in *Astronomy & Astrophysics* (Celiz et al. 2025)
- 2021–2023 I was a mentor for two years in the Colombian Network of Students in Astronomy (RECA) mentoring program. During this time, I supervised the finishing process of two undergraduate Colombian students, and their admission to Master or PhD programs.
- 2018–2023 Local Organizing Committee, Night of the Museums, Astronomical Observatory of Córdoba, Argentina

Service

- 2024–Present Referee for: *The Astrophysical Journal (ApJ)*, *Astronomy & Astrophysics (A&A)*, *Monthly Notices of the Royal Astronomical Society (MNRAS)*, *Nature Communications (Nat. Commun)*, *The Open Journal of Astrophysics (OJAp)*
- 2023 Local Organizing Committee, Conference: Galaxy Formation & Evolution in Southern California (GalFRESKA) meeting. UC Riverside-California
- 2016 Local Organizing Committee, Conference: Diálogo hacia una formación integral en ciencias, Universidad Antonio Nariño (UAN)

Skills

- Programming: Python, Fortran, Bash, Git, LaTeX, SQL
- Languages: Spanish (native), English (fluent)
- Computing and work environments: Linux/Unix, HPC (clusters, SLURM), Git/GitHub
- Teaching: Course design, student mentoring, inclusive pedagogy

References

- Dr. Laura V. Sales Associate Professor - University of California, Riverside
Physics & Astronomy Department
lsales@ucr.edu
- Dr. Mario G. Abadi Associate Professor - National University of Cordoba, Argentina
FAMAF and PI CONICET, Argentina
mario.abadi@unc.edu.ar
- Dr. Julio Navarro Associate Professor - University of Victoria, Canada
Physics & Astronomy Department
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Publications as first author (8)

- Benavides, José A.**, Sales, L. V., Navarro, J. F., White, S. D. M., Frenk, C. S., Oman, K. A., & Cole, S. (2026). The Abundance of Thin Dwarf Galaxies: a Challenge for Cosmological Simulations. *The Open Journal of Astrophysics*, 9, 62091. <https://doi.org/10.33232/001c.162091>
- Benavides, José A.**, Sales, L. V., Wetzel, A., Moreno, J., Feldmann, R., Mercado, F. J., Bullock, J. S., Hopkins, P. F., Faucher-Giguère, C.-A., Stern, J., Wheeler, C., & Kereš, D. (2025). Disks no more: the morphology of low-mass simulated galaxies in FIREbox. *Monthly Notices of the Royal Astronomical Society*, 544(4), 4651–4664. <https://doi.org/10.1093/mnras/staf1847>
- Benavides, José A.**, Navarro, J. F., Sales, L. V., Pérez, I., & Bidaran, B. (2025). The Environmental Quenching Mechanisms of Field Dwarf Galaxies. *The Astrophysical Journal*, 985(1), Article 86, 86. <https://doi.org/10.3847/1538-4357/adced0>
- Benavides, José A.**, Sales, L. V., Abadi, M. G., Vogelsberger, M., Marinacci, F., & Hernquist, L. (2024). Large Dark Matter Content and Steep Metallicity Profile Predicted for Ultradiffuse Galaxies Formed in High-spin Halos. *The Astrophysical Journal*, 977(2), Article 169, 169. <https://doi.org/10.3847/1538-4357/ad8de8>
- Benavides, José A.**, Sales, L. V., Abadi, M. G., Marinacci, F., Vogelsberger, M., & Hernquist, L. (2023). Origin and evolution of ultradiffuse galaxies in different environments. *Monthly Notices of the Royal Astronomical Society*, 522(1), 1033–1048. <https://doi.org/10.1093/mnras/stad1053>
- Benavides, José A.**, Biviano, A., & Abadi, M. G. (2023). DS+: A method for the identification of cluster substructures. *Astronomy & Astrophysics*, 669, Article A147, A147. <https://doi.org/10.1051/0004-6361/202245422>
- Benavides, José A.**, Sales, L. V., Abadi, M. G., Pillepich, A., Nelson, D., Marinacci, F., Cooper, M., Pakmor, R., Torrey, P., Vogelsberger, M., & Hernquist, L. (2021). Quiescent ultra-diffuse galaxies in the field originating from backsplash orbits. *Nature Astronomy*, 5, 1255–1260. <https://doi.org/10.1038/s41550-021-01458-1>
- Benavides, José A.**, Sales, L. V., & Abadi, M. G. (2020). Accretion of galaxy groups into galaxy clusters. *Monthly Notices of the Royal Astronomical Society*, 498(3), 3852–3862. <https://doi.org/10.1093/mnras/staa2636>

Publications as coauthor (14)

- Sadhu, P., Sales, L. V., Navarro, J. F., Errani, R., **Benavides, José A.**, & Peng, E. W. (2026). The abundance and radial distribution of faint and ultra-faint dwarfs in galaxy clusters. *Submitted to MNRAS, arXiv e-prints*, Article arXiv:2604.22920, arXiv:2604.22920.
- Rodriguez, S., Yaryura, Y., **Benavides, José A.**, Garcia Lambas, D., Pedrosa, S., Baravalle, L. D., Ceccarelli, L., Luparello, H. E., Bignone, L., & Galaz, G. (2026). The connection between surface brightness and satellite systems for central galaxies through IllustrisTNG. *Astronomy & Astrophysics*, 707, Article A372, A372. <https://doi.org/10.1051/0004-6361/202557447>
- Goyal, C., Ponnada, S. B., Hopkins, P. F., Wellons, S., **Benavides, José A.**, & Su, K.-Y. (2026). Effects of Varied Cosmic Ray Feedback from AGN on Massive Galaxy Properties. *Publications of the Astronomical Society of the Pacific*, 138(3), Article 034102, 034102. <https://doi.org/10.1088/1538-3873/ae4a4c>
- Celiz, B. M., **Benavides, José A.**, & Abadi, M. G. (2025). Compact groups of galaxies in the TNG100 simulation. *Astronomy & Astrophysics*, 702, Article A22, A22. <https://doi.org/10.1051/0004-6361/202555375>
- Mancera Piña, P. E., Read, J. I., Kim, S., Marasco, A., **Benavides, José A.**, Glowacki, M., Pezzulli, G., & Lagos, C. d. P. (2025). The galaxy-halo connection of disc galaxies over six orders of magnitude in stellar mass. *Astronomy & Astrophysics*, 699, Article A311, A311. <https://doi.org/10.1051/0004-6361/202554381>

- Arjona-Gálvez, E., Cardona-Barrero, S., Grand, R. J. J., Di Cintio, A., Dalla Vecchia, C., **Benavides, José A.**, Macciò, A. V., Libeskind, N., & Knebe, A. (2025). A physically motivated galaxy size definition across different state-of-the-art hydrodynamical simulations. *Astronomy & Astrophysics*, 699, Article A301, A301. <https://doi.org/10.1051/0004-6361/202554290>
- Mercado, F. J., Moreno, J., Feldmann, R., Zeender, M., **Benavides, José A.**, Piotrowska, J. M., Klein, C., Wheeler, C., Necib, L., Bullock, J. S., & Hopkins, P. F. (2025). Effects of Galactic Environment on Size and Dark Matter Content in Low-mass Galaxies. *The Astrophysical Journal*, 983(2), Article 93, 93. <https://doi.org/10.3847/1538-4357/adbf07>
- Costa, A. P., Ribeiro, A. L. B., de Carvalho, R. R., & **Benavides, J A.** (2024). Boosting the evolutionary picture of galaxy clusters: a case study at intermediate-redshift of CI 0024+17 and MS 0451-03. *Monthly Notices of the Royal Astronomical Society*, 535(2), 1348–1363. <https://doi.org/10.1093/mnras/stae2410>
- Klein, C., Bullock, J. S., Moreno, J., Mercado, F. J., Hopkins, P. F., Cochrane, R. K., & **Benavides, José A.** (2024). Size-mass relations for simulated low-mass galaxies: mock imaging versus intrinsic properties. *Monthly Notices of the Royal Astronomical Society*, 532(1), 538–548. <https://doi.org/10.1093/mnras/stae1505>
- Sazonova, E., ..., **Benavides, José A.**, & ... (2024). RMS asymmetry: a robust metric of galaxy shapes in images with varied depth and resolution. *The Open Journal of Astrophysics*, 7, Article 77, 77. <https://doi.org/10.33232/001c.123524>
- Malek, K., ..., **Benavides, J. A.**, & ... (2024). Attenuation proxy hidden in surface brightness-colour diagrams. A new strategy for the LSST era. *Astronomy & Astrophysics*, 684, Article A30, A30. <https://doi.org/10.1051/0004-6361/202348432>
- Doppel, J. E., Sales, L. V., **Benavides, José A.**, Toloba, E., Peng, E. W., Nelson, D., & Navarro, J. F. (2024). Imposters among us: globular cluster kinematics and the halo mass of ultra-diffuse galaxies in clusters. *Monthly Notices of the Royal Astronomical Society*. <https://doi.org/10.1093/mnras/stae647>
- Martin, G., ..., **Benavides, J. A.**, & ... (2022). Preparing for low surface brightness science with the Vera C. Rubin Observatory: Characterization of tidal features from mock images. *Monthly Notices of the Royal Astronomical Society*, 513(1), 1459–1487. <https://doi.org/10.1093/mnras/stac1003>
- Ferrero, I., Navarro, J. F., Abadi, M. G., **Benavides, José A.**, & Mast, D. (2021). A unified scenario for the origin of spiral and elliptical galaxy structural scaling laws. *Astronomy & Astrophysics*, 648, Article A124, A124. <https://doi.org/10.1051/0004-6361/202039839>

Others publications and ArXiv

- Wetzel, A., Samuel, J., Gandhi, P. J., Ponnada, S. B., Su, K.-Y., Arora, A., Angles-Alcazar, D., Hayward, C. C., Sanderson, R. E., Feldmann, R., Cochrane, R., Nikakhtar, F., Panithanpaisal, N., **Benavides, Jose A.**, Pandya, V., Grudic, M., Hummels, C., Gurvich, A. B., Hafen, Z., ... Hopkins, P. F. (2025). Second public data release of the FIRE-2 cosmological zoom-in simulations of galaxy formation. *arXiv e-prints*, Article arXiv:2508.06608, arXiv:2508.06608.
- Celiz, B. M., **Benavides, J. A.**, & Abadi, M. G. (2023). Formación de grupos compactos de galaxias en simulaciones cosmológicas hidrodinámicas. *Boletín de la Asociación Argentina de Astronomía La Plata Argentina*, 64, 229–231.
- Benavides, J. A.** & Abadi, M. G. (2021). Ensamblaje de cúmulos de galaxias por eventos de acreción de grupos de galaxias. *Boletín de la Asociación Argentina de Astronomía La Plata Argentina*, 62, 171–173.
- Daza, I., Alfaro, I. G., **Benavides, J. A.**, Lares, M., Santucho, M. V., Cabral, J. B., O'Mill, A. L., Rodríguez, F., & Koraj, M. (2021). PISCIS: Platform for Interactive Search and Citizen Science. *Boletín de la Asociación Argentina de Astronomía La Plata Argentina*, 62, 310–312.

- Benavides, J. A.**, Abadi, M. G., & Sales, L. V. (2022). El origen de las galaxias Ultra-Difusas apagadas en el campo. *Boletín de la Asociación Argentina de Astronomía La Plata Argentina*, 63, 208–210.
- Benavides, Jose A.**, Chalela, M., Cabral, J. B., Sánchez, B. O., & Gurovich, S. (2021). SCORPIO, a package for the visualization of galaxy pairs. *arXiv e-prints*, Article arXiv:2111.01260, arXiv:2111.01260.
- Benavides, José** & Casas-Miranda, R. A. (2018). Simulaciones numéricas de la formación del disco de satélites de la vía láctea a partir de la acreción de asociaciones de galaxias enanas. *Revista de la Academia Colombiana de Ciencias Exactas, Físicas y Naturales*, 42(162), 32–40. <https://doi.org/10.18257/raccefyn.561>
- Benavides, J.** & Casas-Miranda, R. A. (2018). Infall of Associations of Dwarf Galaxies into the Milky Way Halo. *Revista Mexicana de Astronomía y Astrofísica Conference Series*, 50, 13–14.

Selected Conference

- Benavides, J.** & Sales, L. (2025). *10 Years of Ultra-Diffuse Galaxies, past perspectives and future prospects. Benasque, Spain.*
- Benavides, J.**, Navarro, J., & Sales, L. (2024). *Mind the Gap: Galaxies and Large-Scale Structure. Córdoba, Argentina.*
- Benavides, J.**, Sales, L., & Abadi, M. (2024). *Workshop: Dwarf Galaxies, Star Clusters, and Streams in the LSST Era. Chicago, USA.*
- Benavides, J.**, Sales, L., & Abadi, M. (2023a). *4th Regional Extragalactic Astronomy Meeting: 60 Years Of The Sersic Law. Córdoba, Argentina.*
- Benavides, J.**, Sales, L., & Abadi, M. (2023b). *XVII Latin American Regional IAU Meeting. Montevideo, Uruguay.*
- Benavides, J.** & Sales, L. (2023). *Hiking through the unexplored Universe: the sunrise of ultra diffuse galaxies. Sexten, Italy.*
- Benavides, J.**, Abadi, M., & Sales, L. (2022a). *Special Seminar, hosted by the Galaxy Clusters group. Institute for Fundamental Physics of the Universe (IFPU), Trieste-Italy.*
- Benavides, J.**, Abadi, M., & Sales, L. (2022b). *XI Friends of Friends Meeting – Linking astronomy and friends. Universidad Nacional de Córdoba, Córdoba-Argentina.*
- Benavides, J.** & Abadi, M. (2021). *X Friends of Friends Meeting – Linking astronomy and friends. Universidad Nacional de Córdoba, Córdoba-Argentina.*
- Benavides, J.** (2017). *Encuentro Latinoamericano eCiencia. San Jose-Costa Rica.*